VII SEMESTER (PROFESSIONAL ELECTIVE-III)	L	T	P	C
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20EC7E03 - SPEECH PROCESSING				

COURSE OUTCOMES:
Upon completion of the course, students will be able to:
CO1: Illustrate the speech production system and describe the time domain methods.(K1,K3)
CO2: Describe the frequency domain methods in speech processing (K1)
CO3: Explain the linear predictive analysis of speech processing.(K2)
CO4: Develop various speech enhancement techniques.(K3)
CO5: Understanding the basics of speech production and perception(K4)

UNIT- I: SPEECH PRODUCTION:
Speech signal; Speech Production process: Lungs, Larynx and Vocal folds, Vocal tract; Acoustic Phonetics: Vowels, Diphthongs, Semi vowels, Nasals, Unvoiced fricatives, Voiced fricatives, Voiced and unvoiced stops; Acoustic theory of speech production; Digital model for speech signals.

UNIT-II: TIME DOMAIN METHODS FOR SPEECH PROCESSING:
Time domain parameters of Speech signal, Methods for extracting the speech parameters (Energy, Average Magnitude, Zero crossing Rate), Silence Discrimination using Zero crossing Rate and energy, Short Time Auto Correlation Function, Pitch period estimation using Auto Correlation Function.

UNIT-III : FREQUENCY DOMAIN METHODS FOR SPEECH PROCESSING:
Short-Time Fourier Transform (STFT), Sampling the STFT in Time and Frequency, The Speech Spectrogram, homomorphic speech analysis: homomorphic systems for convolution, Definition of the Cepstrum and complex Cepstrums, pitch extraction using homomorphic speech processing.

UNIT-IV: LINEAR PREDICTIVE ANALYSIS OF SPEECH:
Linear prediction of speech, auto correlation, formulation of Linear prediction coding equations, Solution of Linear prediction coding equations, Levinson Durbin recursion, Application of Linear prediction coding parameters: Pitch detection using Linear prediction coding parameters, Deriving acoustic parameters PLPs, LPCCs, and MFCCs from LPCs

UNIT-V: SPEECH ENHANCEMENT:
Nature of Interfering Sounds; Speech Enhancement (SE) Techniques: Basic principles of Spectral Subtraction; Wiener Filtering; Wiener filtering for noise reduction; Statistical-Model-based method: Maximum-likelihood estimator for speech enhancement; Applications of speech enhancement.

Text Books
1. Lawrence R. Rabiner and Ronald W. Schafer, “Introduction to Digital Speech Processing” now Publishers Inc. Hanover, USA, 2007.
2. Thomas F. Quateri, “Discrete Time Speech Signal Processing: Principles and Practice”, 1st Ed., PE, 2008.
3. Philipos C. Loizou, “Speech Enhancement” 2ndEdition, CRC Press, Taylor & Francis Group, 2013

References
1. Douglas O Shaughnessy, “Speech Communications Human and Machine” 2nd Edition, IEEE Press, 2000.
2. Speech and Audio Signal Processing, Processing and Perception of Speech and Music- Ben Gold and Nelson Morgan, Wiley- India Edition, 2006.